NEURALINK – Cognitive Audio Generation

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First of all Thank You Elon for actually beginning what I've been thinking about for 30 years.

We are witnessing an unprecedented convergence in consumer interfaces and cognitive computing. Brain-Computer Interfaces (BCIs) are rapidly transitioning from clinical, mobility-assistive trials to mainstream consumer realities. Efforts at present focus on restoring sight or mobility, while tech giants are quietly building cognitive monitoring infrastructure—such as Meta harvesting neural signals via earbud sensors and glasses, and China making massive, state-backed BCI advancements. High-dimensional generative models have revolutionized image generation, the way we communicate and create and transform audio digitally. However, humans are bottlenecked by physical interfaces; we cannot express complex acoustic ideas using text prompts or physical sliders.

However there is a disconnect in the high-dimensional inverse mapping between the completely different dynamic systems of the brain and generative AI with severe structurally dissimilar modalities. We are on the precipice of cognitive creation, yet there is a conspicuous lack of dialogue or active push from professional audio for direct cognitive audio capabilities via BCI or "Cognitive Audio Generation" CAG. As the neural computing transition accelerates over the next decade, the creative, communication, and security landscapes will drastically change and split cleanly between those who pioneer cognitive sound synthesis and those who rely on legacy paradigms.

The Winners Will Be The Creators of the Foundational Cognitive AV Codec

Pioneers who create the fundamental audio (and visual) generation layer will establish the foundational neural Audio Visual Codec. Ffmpeg for generation via BCI leveraging AI. By licensing this core technology to every consumer BCI provider and hardware platform, they will unlock massive, as yet un-forcasted revenue streams. These leaders will define the entertainment, spatial experiences, and advances in secure systems for generations to come.

The Losers: Competitors & Legacy Passive Consumers

Those who treat BCI solely as a passive consumer consumption medium (AR/Meta) or as a brilliant but limited medical rehabilitation tool (at least until we go cyborg) will become end users. Legacy software and hardware DAWs (Digital Audio Workstations) will face terminal decline or be forced to adapt and integrate, unable to compete with real-time thought-speed workflows, leaving late adapters trapped behind low-bandwidth interfaces. I think even those who want to use vintage software for the feels,will gain a significant advantage by controlling their use with BCI.

The actual problem which I would like to solve by coming to work with Neuralink is this… And I have wanted this since I began writing electronic music at age 13, 30 years ago.

I have always wanted to just plug an audio cable into my head and dump the output straight into a DigitalAudio Workstation.

Enabling complex sound generation, musical writing via brain BCI and being able to control the software I was using by thought. To transform and effect audio in diverse ways that only I could hear. There was and still is a bandwidth problem and more importantly the tweaks, effects finesse and the unique ways in which I want to transform or generate a sound are severely limited by the capabilities of the software/hardware tools available now and historically. The problem in my experience is only made more frustrating with AI audio generation and natural language tools.

My Promised Land is a world where the human mind communicates directly in coherent wave formats—achieving true creative synesthesia. By plugging a cognitive audio connection directly into software, we eliminate the physical delays of keyboards, mouse clicks, and manual dial tweaks.

In this future state, a creator could seamlessly generate complex, finely defined auditory parameters from imagination directly into high-fidelity acoustics.

Bridging the gap to the Promised Land requires solving highly complex computational, mathematical, and neurological challenges.

Manifold Learning and Generative Topology

A human cannot consciously modulate millions of individual neural channels simultaneously. A core engine would need to bypass this by utilizing manifold learning and generative topology to compress the information-theoretic gap, shifting low bandwidth neural control signals into high-bandwidth acoustic generation through a shared mathematical manifold.

We need to solve the Inverse Electroencephalography Problem

By creating a system that uses specialized neural decoding structures to reconstruct active creative intent from electrical micro voltages on the scalp (or intracortically) ***V(t) → I(t)***, bypassing the traditional physical noise and bone attenuation issues.

> *"Now this, gentlemen, is the catalyst. A neurological synaptic transfer system. It not only maps the mind it sends the signal to another party." The Cell - 2000 - New Line Cinema*

Cognitive Language Mapping of Sound

Instead of lengthy, tedious training sessions for every individual user, the model would need to tap into clearly defined brain regions dedicated to complex language, environmental sound processing and executive function. We should leverage emerging, highly targeted EEG datasets of creative processes and imagination and license the neural signals harvested by Meta Glasses and Apple Ear Pods (US20230225659 - 20.07.2023) and as many other Consumer Electronics providers that harvest relevant data so we have a head start datawise for AI to use when mapping for a system natively optimized for audio.

A hypothetical non-creative use case could be something like:

CAG tools could act as a unique one time encryption keyphrase. For instance, in secure military comms where voice communication is needed — a generated piece of audio could be generated by user 1, mathematical transforms applied by the BCI, transmitted via satcom, received by user 2 their BCI performs the inverse math transform, the key matched to a stored memory of user 1's unique audio. Such as real-time tac-com between an A-10 Warthog pilot and ground troops calling in target coordinates — It could be custom encrypted to contain complex formant data unique to a user's voice or a voice they imagine using CAG.

As a completely off the top of my head estimate of a timeline of the R&D path I would imagine if significant resources are brought to bear on the problem that a realistic time frame would be 5-15 years. But I am just making that up as I have no ability to perform fundamental or statistical analysis for forecasting due to lack of access to developmental data which is comparable. But Neuralink has been 10 years so far and we are still a ways off an accurate simulation of a brain (Human no)(Fruit fly yes) That estimate is from my attempt at Kurzweilian exponential improvement graphing.

Here are my reasons I know I can help Neuralink be first to market with Cognitive Audio Generation with a first focus on Audio.

I want and need to contribute my cognition and skills to something concrete, meaningful and to the measurable benefit of man, earth and its other lifeforms.

- I have come up with solutions to technical problems creatively in the creative sphere.

An excellent example of which is this…

When I was 14 years old I had a friend who owned a very new set of the legendary Technics SL1200 turntables upon which we were learning to DJ. He knew I made electronic music because I wouldn't shut up about it and was quite experienced with such having been doing so for about a year and a half by that point in time.

He asked me “If I were going to play an MP3 off the computer but control it with a piece of vinyl so you could scratch and beatmix, how would I do that?” I responded instantly.

“Well because Digital Audio is basically a whole bunch of little slices of data. What you could do is press a consistently rising tone to vinyl. Map the start of the tone (Low) to the beginning of the MP3 and the end of the tone (High) to the end of the MP3, Run an output off the mixer into the PC where the software would map frequency to digital audio slice sequentially low to high start to finish. Then have the audio play off the computer back into the mixer. You could control the speed and pitch of sequential audio slices as they play, based upon the location of the needle on the tone vinyl. I'm not 100% sure if he was the one that capitalised on that exact idea but about 6 months later Serato the first DJ software with that exact solution Serato NoiseMap™, which was founded in New Zealand.

II've got my sneaking suspicions that it was him and his older brother. Because they all of sudden moved to Asia and got very comfy in places where audio electronics are made.
Technically the system uses a maximal-length pseudo-random bit sequence generated by a Linear Feedback Shift Register (LFSR).

But when you listen to the audio signal it is a constantly rising tone. Patent (US6266003) Either way I think it's a good example of how I got the theory of a foundationally new technology which transformed an industry exactly right without having to think about it twice, using knowledge, logic and intuition.

Now I haven't been in touch with this friend in a while, but if you would like to independently verify the story, I would be happy to privately provide information which would enable you to track them down independently and ask him if he asked me that question and if that was my response. I'm not sure if you would remember but the offers there in the interest of trust.

- I also have an inherent ability to correctly identify paths in technology by using logic and being tenacious with little details.

For example I tracked down Patent (US9685513B2): Held by The U.S. Department of the Navy, following only my own logical extrapolation of current technology. I'm sure your starlink benefits from this one via Marvell. That was me trying to come up with a better material than silicone for semiconductors, knowing diamond was awesome with heat and was also a crystal researching to see if anybody had tried to make chips out of it and researching the created diamond industry. Found the Navy info via an obscure article got up to speed with MPCVD and 2D Graphene uses and producers all for the sake of curiosity. It has benefited me this year stocks-wise by consistently starting from scratch with the same research logic and topic. You never know what new information is out there and if you start from where you left off last time you provide a massive filter to potential new information.

I am so bold as to presume to surmise a possible connection between Elon and Echostar/Conundrum Wireless and possible use of the Guam AWS-3 Licences purchased on June 2, 2026 in Auction 113. Guam, a not so quiet little island, has the potential for a very specific use case. I do believe saying so here is about as far as I will push speculation as rudeness is not an admirable trait. Though bluntness and demonstration of ability are both traits exhibited by Elon.

The Exact Way I Would Like to Contribute at Neuralink.

For starters I can't honestly see myself excelling anywhere but in the fold of Elon's companies, in a variety of roles.

For Cognitive Audio Generation, R&D and Implementation with Neuralink.

- IDENTIFICATION
- MAPPING SPECIFICATIONS
- THE GATHERING AND COORDINATION OF HUMAN SOURCES FOR AI TRAINING ● PLANNING AND ASSIST TO THE DEV AND ENGINEERING TEAM
- CONTRIBUTING TO THE GUIDANCE AND SPIRIT JOURNEY OF CAG
- HELPING WITH R&D AND DEV PLANNING TO MEET SPECIFICATIONS
- WITH MEDIA NARRATIVE AND DRUMMING UP CONSUMER INTEREST AND INTEREST FROM INDUSTRY
- CONSULTING ON POTENTIAL STRATEGIC INDUSTRY AND INDIVIDUAL AUDIO PROFESSIONAL PARTNERSHIPS.

By identifying who to speak to so we can identify the broad wants and deliverables musicians and sound engineers do consider as fundamental to their craft and the Mapping onto Specifications for a more than basic P.O.C Design Brief. The identification of a broad set of contributing audiophiles who want to help us train AI to solve this unique but fundamental use case for BCI. Being at the disposal of the rest of the BCI TEAM.

Providing a "this is what I would want" and "this is what they said they want" level of guidance and interpretation of functionality and deliverable rules identified by industry leading hardware/software providers such as AVID/DIGIDESIGN, by musicians,by sound engineers for game and film and other devs and audiophiles that we engage with in this process.

I'm pretty good at business administration and coordination having experience operating my own companies since 2006. Whilst professionally engaging with pro-audio from multiple frames of reference including DJ performance, extensive roadie work, film sound and post-production process customization and optimization of workflows via custom code integrating with top-tier industry standard software and 40years of audio production / song writing.

Guaranteed value I know I can provide is having used so many different audio applications and hardware that I cannot think of them all and listing would be a task similar in scope to listing every different album

I've listened to. and the entire time since 1995 I have been imagining how I really wanted things to sound in my head and not having an output for that I systematically began thinking about BCI's. Practicing sound generation and effecting of sounds in my head in such a way as my thoughts are coherent and structured and then trying to match that with software. I am now 44 and am seeing reality being formatted before my eyes by the correct and concerted effort of Elon. In ways I have been fantasizing about. It provides me with a consistent stream of enthusiasm and drive to get involved as we evolve.

Impact

The immediate impact of us beginning to solve the CAG problem, I think would be to generate, I would estimate, some significant interest about the technological timeline and usefulness of BCI's to the audio industry as a whole and surge in enthusiasm from individuals who are closely following and using AI content generation capabilities at present.

Personally, my enthusiasm for the augmented ability of CAG and its implementation is such that I would be wanting to contribute as much as possible and would tirelessly be thinking about it and more broadly other systems and use cases which are similar. How what we are doing could be applied to different Cognitive Generation Cases and how they could be approached during/via its development.

Issues and Flaws

The critical issue blocking progress to CAG in my view is a focus on helping those with limited mobility or sight first and then looking to other use cases later. Whilst audio generation has been as far as I can tell completely ignored. Whereas there are those who cannot speak, who used to be able to speak or have been mute entirely, there is also a lack of discourse on the topic or push for capability from audio professionals for creative industries. It is going to be a much harder task than moving a cursor on screen with a little training, this is going to require at least a partial mapping of the brain and having that map be applicable rather broadly to different people. The engineering of a codec which enables encoding/decoding/transcoding and in such a way as it's pretty much instantaneous.

These all provide significant hurdles, which we need to get to work on now. We will find more I am sure so we best get started on the list asap.

Why?

I can see this being a very significant revenue stream once v1.0 Pro is launched, if implemented even barely possibly well. I can also see there being a rapid jump in people wanting to customize for different specific use cases or tooling, exactly similar to the rapid rise in MCP Server tooling and customization.

Why am I approaching you? Why does it need to be fixed now? I can only think of the Cryptonomicon and have you read Snowcrash by Neil Stephenson? An ancient virus spreads verbally across the Metaverse. There is a security conversation which does need to be had regarding CAG and similar capabilities.

With no clear leader thinking about CAG the company that does so we'll have a unique opportunity for foundational codec growth and associated licensing to other BCI providers until they develop their own. But first to market dictates, quality, technical standards and will have more opportunity to use the tech for good.

Solved

This problem opens up a world of possibilities and with the transposition of audio generation, transformation and interpretation, which fundamentally is an exercise in structuring waves into coherent output waves that are meaningful and useful, and closely aligned with imagined output, there will be the ability to via a type of synesthesia apply the developmental framework used to generate art, images, movies, imagine the structure of electromagnetic fields and dictate via what we've learned into other coherent wave formats. For example looking at unprocessed raw film footage on screen of a scene and imagining how you want it to look and having that be output properly and with detail that you imagine , effect and add elements to the composition, change coloring, add effects and transitions. All start as minute waves in the brain and we are trying to change a scene made up of light and colors in waves. Our development of CAG could provide a framework for other exercises in coherent wave formatting.

Difficulties

I think this is going to be quite a difficult thing to solve and I would imagine the timeline could be around 15 years if we started working on it now, with some significant resources. The reason I think that it would take that long is because you kind of can't hear things in your head, you kind of imagine texture and you kind of are unable to transfer something like the feeling of your body vibrating to a certain bass frequency for example or the edginess of a string section with just a touch of distortion on them and then a slight phaser or specific roominess of a reverb. These are difficult and very finely defined digitally. But are contextual and impressions in our head. Diverse properties which have different results such as the amount of delay time in a flanger and that flanger being applied to two saw waves which are offset from each by a certain amount of samples to have the effect of unison in a certain way, it is such a broadly scoped multifaceted, hard to define but easy to imagine problem. A simple solution to a problem of which I do not think much thought has been put into so far , someone needs to begin thinking about this now. Beginning planning, preparation, etc.

I could be wrong and it could be easy. At the very least I do know that being able to think communications between an A10 Warthog pilot and ground troops could be uniquely encrypted using techniques surrounding audio generation and imagination via a BCI just as an example.

I want to find out how difficult this problem really is and the only way to do that is to be speaking directly to, over time maybe sometimes over a beer after hours with the engineers and dev team that are getting Neuralink off the ground. To have the opportunity with the backing of a known respected name which would make it an easy thing to get an hour to of Hans Zimmer's time or go and speak with Iggy Pop before he dies and ask him very carefully about how he imagines these things should/could be. To record them imagining audio, to be involved in the comparative use of such data to build a model. I definitely can't do those things alone but together it's doable. A task we should do.

I have developed a network of professional musicians and am personally friends with Grammy award film sound engineers and lifelong musicians, specialised on a certain instrument, culturally via family tradition. I can easily contact cutting edge electronic music producers who literally built genres up from the roots in the early 90's and are perfectionists of their craft.

Finally to personally blow my own trumpet. As far as I know I was the first person in New Zealand to begin making Drum and Bass.

Although in my hundreds of Tunes I have produced I have never released any music for my own personal reasons and issues with the morals and culture of the music industry. The use of Cognitive Audio Generation by Neuralink BCI decentralizes and personalizes music in a way no other thing can and ultimately puts the listener, the fan, the amateur and the pro all on the same level playing field where imagination is quite actually the only limitation.

Musical influences include extreme perfectionists such as Ed Rush and Optical - Virus Records, the consistently beautiful guitarist Paola Hermosin - and for motivation in the morning the second song in my get up and code playlist is Wilson Phillips - Hold On.

"Ride the Tiger" - Dio